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(1) Research Program/Project/Study Title: Development of Hydro-Solar Floating Generator	
Proponent Name:	Jayvee D. Crespo, MAT–Math - Instructor 1
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	Felix F. Facto, EdD - Professor 4, Ma. Luisa P. Balatayo, Asso. Prof. 5, George D. Daliva, MAIEd-IA, Instructor 1
Campus/Satellite College/Address:	CapSU – Main Campus, Roxas City
Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC
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		Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center						
(4) Site of implementation: Capiz State University, Main Campus								
Municipality/City:		Roxas City						
Province:		Capiz						
Region:		VI						
(5) Classification <table border="0"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☒ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☒ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:							
☐ Basic	☒ Pilot Testing							
☐ Applied	☐ Tech. Promotion/Commercialization							
(7) Sector/Commodity: Agricultural		(8) Discipline/Field of Specialization: Electrical Technology						
(9) Rationale: Hydro-Solar Floating Generator is a device that will: - Respond to the call of DIT RDE thrust and priorities specifically on clean energy/ alternative energy. - Provide perpetual electrical energy source thru solar and tidal water of ponds and canals. - Promote and enhance alternative energy utilization, and environmental protection.								
(9-A) Project Description: This hydro-solar floating generator is adaptable to the water level of ponds, may it low or high tide water. The direction of rotation does not limit only on forward but both reverse and forward directions. This tidal water energy source has a solar tandem to ensure a stable electrical energy source. The possible load that this generator could operate is Light Emitting Diode Lamps or LED lights. This is ideal for coastal residents for it utilizes a combined solar and sea								

water. Utilizing this generator does not cause adverse effects against environment or pollution that destroy our natural resources.

(10) Objectives/Statement of the Problem:

Generally, this study will be conducted in order to develop a hydro-solar floating generator. Specifically, this study aims to determine the following under normal atmospheric conditions:

1. How long will a hydro-solar floating generator:
 - a) fully charge its seven (7) plates storage battery in terms of time in hour; and
 - b) operates a ten (10) watts LED lamps, and three (3) cell phone battery chargers?
2. What is the acceptability level of a hydro-solar floating generator as evaluated by its target end users in terms of:
 - (a) Design
 - (b) Composition
 - (c) Performance

(11) Review of Literature/Prior Art Search

Tidal power is entirely predictable unlike wind and solar energy. It therefore, provides the potential to be used as a base-load power supply that is immune from climatic conditions and anthropogenic demands. For example, the amount of power generated is strongly related to the size of the tidal range. The output varies with the square of the tidal range, i.e., if a tidal range of x gives a power output of y , then a tidal range of $10x$ will give a power output of $100y$. The power output is also directly related to the area of the impoundment structure that, in-turn, dictates the volume of water passing through the turbine during each generating phase. Tidal power generating equipment is also very efficient compared to other modes of power generation and, in general, the bigger the power generating equipment, the higher the efficiency. Continuous power generation, that matches electricity demand, is an important characteristic of electricity supply. Generally, this is achieved through the continuous running of base-load generation equipment supplemented with peak-load generation equipment during peak periods. Nuclear power plants cannot be shut down quickly and are operated hours per day and, therefore, nuclear power is a base load type of power. Fossil fuels are somewhat easier to shut down but typically they vent the steam to the atmosphere when demand for electricity declines at night. On the other hand, tidal power is a power source that can be shut down quickly and restored quickly without major losses in efficiency.

An Example of Methodology for the Floating Wave Power Vessel to Convert Wave Energy into Electricity is Floating Wave Power Vessel (FWPV) based on ship construction, will be anchored in 50-80 m depth of water. It will be designed to have a maximum power output of 1.5 MW, producing about 5.2 million kWh per year. This design functions by capturing the water from waves that run up its sloping front face. The captured water is returned to the sea via a standard Kaplan hydro-electric turbine. In many respects this is a floating Tapchan. International Journal of Scientific & Engineering Research Volume 3, Issue 12, December-2012.

Furthermore, this Floating Wave Power Vessel loaded Tidal Stream Generator to convert Wave Energy into Electricity. Material selections were made using Engineering Business Ltd.'s (EB) in house knowledge of building subsea machines. The timescale of the project and cost limitations also had a large influence on specification. For example, a commercial machine designed to be in the water for many years would have a greater number of stainless steel

components to resist corrosion and allow for easy maintenance over its lifetime. However, a compromise with cost had to be made in this project and in many places painted steel parts could be used at a fraction of the equivalent stainless steel cost. Due to the size of the machine the design allowed for the machine to be broken down easily into key parts for transportation by barge and re-assembly on site. Preliminary designs had been for a twin hydroplane machine. During the design process this was simplified to a single hydroplane design to ease manufacture and reduce cost. Parametric modeling of performance and cost had shown that a single hydroplane could be configured such that it had similar overall performance to that of the optimized twin hydroplane machine developed, Kamrul Alam Khan et al (2012).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology

Description of the Device

The materials used in fabricating hydro-solar floating generator includes: 200volt-ampere modified generator, 8-blade water wheel with 15x30cm blade dimension. 3-level speed chain multipliers, base floater. This tidal source is tandem with 100-watt solar panel and seven (7) plates battery for storing electrical energy for continuous operation.



Test Procedures

The hydro-solar floating generator is deliberately designed to operate at a minimal water of low tide and high tide flowing in and out of the ponds or canal. To check the reliability of the hydro-solar floating generator as to its capability of generating electricity continuously, it must be placed in the entrance of the pond or canal where water is enough to drive the blade. While the generator is operating, checked if voltage is generated using the voltmeter. Knowing that the voltage is generated, the next step is to plug the seven plates battery and observe until it reaches fully charged. Repeat the same procedure to ensure data consistency for ten (10) times. When battery reaches fully charge, the ten (10) watt LED lamp and three cellphone battery charger started to operate. This procedure will be repeated ten (10) times to check the consistency of the result.

Instrumentation

The data sheet shown below prepared by the researcher will serve as an instrument of the study. This will be used as a record sheet for the generated/ charging and discharging voltage consistency.

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		
- Meal allowance during data gathering		
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		
- Printing		
C. CAPITAL OUTLAY		
Total		
(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/) Kamrul Alam Khan et al (2012) A Study on Tidal Power Conversion for use in Bangladesh		
(20) Capsule Curriculum Vitae (One-page brief C.V. only)		